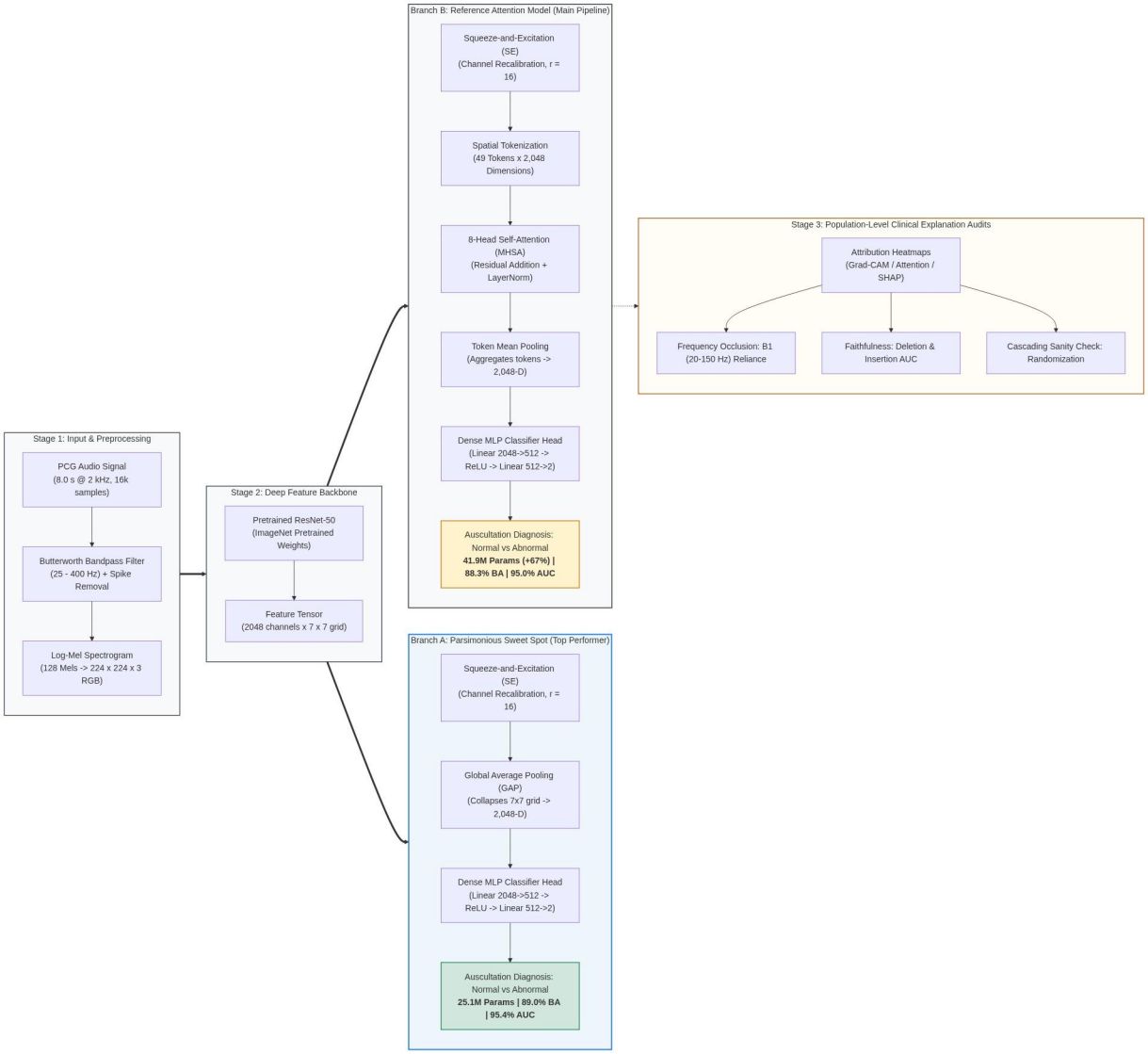

Stage 1: Input & Preprocessing
PCG Audio Signal (8.0 s @ 2 kHz, 16k samples)
Butterworth Bandpass Filter (25 - 400 Hz) + Spike Removal
Log-Mel Spectrogram (128 Mels -> 224 x 224 x 3 RGB)
Stage 2: Deep Feature Backbone
Pretrained ResNet-50 (ImageNet Pretrained Weights)
Feature Tensor (2048 channels x 7 x 7 grid)
Branch B: Reference Attention Model (Main Pipeline)
Squeeze-and-Excitation (SE) (Channel Recalibration, r = 16)
Spatial Tokenization (49 Tokens x 2,048 Dimensions)
8-Head Self-Attention (MHSA) (Residual Addition + LayerNorm)
Token Mean Pooling (Aggregates tokens -> 2,048-D)
Dense MLP Classifier Head (Linear 2048->512 -> ReLU -> Linear 512->2)
Auscultation Diagnosis: Normal vs Abnormal 41.9M Params (+67%) | 88.3% BA | 95.0% AUC
Branch A: Parsimonious Sweet Spot (Top Performer)
Squeeze-and-Excitation (SE) (Channel Recalibration, r = 16)
Global Average Pooling (GAP) (Collapses 7x7 grid -> 2,048-D)
Dense MLP Classifier Head (Linear 2048->512 -> ReLU -> Linear 512->2)
Auscultation Diagnosis: Normal vs Abnormal 25.1M Params | 89.0% BA | 95.4% AUC
Stage 3: Population-Level Clinical Explanation Audits
Attribution Heatmaps (Grad-CAM / Attention / SHAP)
Frequency Occlusion: B1 (20-150 Hz) Reliance
Faithfulness: Deletion & Insertion AUC
Cascading Sanity Check: Randomization